Chem 150 **Answer Key** **Term Test 2A**

1.(**2 marks**) Write a dissociation equation for K_2SO_4 in water.

$K_2SO_4(s) \leftrightarrow 2K^+(aq) + SO_4^{2-}(aq)$

2.(**2 marks**) Write an ionization equation for HNO_3 in water.

$HNO_3\ (l) + H_2O(l) \rightarrow NO_3^-(aq) + H_3O^+(aq)$

3. (**2 marks**) Sodium chloride dissolves spontaneously in water at room temperature. The standard enthalpy for this process NaCl(s) →NaCl(aq) is + 5 kJ/mol. What can you confidently say about the standard entropy for this process?

We know that sodium chloride dissolves spontaneous in water at room temperature (25 °C or 298 K) so ΔG° must be negative, we also know that ΔH° is + 5 $\frac{kJ}{mol}$ so the entropy must be increasing (ΔS° must be positive) which makes sense since we are increasing the randomness (the ions are free to move in x, y and z direction in the salt solution but not in the crystal lattice).

We can be even more specific: TΔS° must be greater than 5 $\frac{kJ}{mol}$ *to make ΔG° negative and since T is 298 K we can say: = TΔS° > ΔH° or (298 K) (ΔS°) >* 5 $\frac{kJ}{mol}$ *or ΔS° >* 16.8 $\frac{J}{mol\ K}$

For the dissolution of sodium chloride in water the standard entropy must increase by at least 16.8 $\frac{J}{mol\ K}$

4. (**3 marks**) Calculate the pH of 0.00112 M $Ca(OH)_2(aq)$

Calcium hydroxide is an ionic compound that has some solubility in water. All the $Ca(OH)_2$ that dissolves dissociates in water making the solution strongly basic.

$Ca(OH)_2(s) \leftrightarrow Ca^{2+}(aq) + 2OH^-(aq)$

0.00112 M $Ca(OH)_2(aq)$ x $\frac{2\ mol\ OH-}{1\ mol\ Ca(OH)_2}$ = 0.00224 M $OH^-(aq)$

pOH = -log[0.00224] = 2.65

pH = 14 – pOH = 11.35

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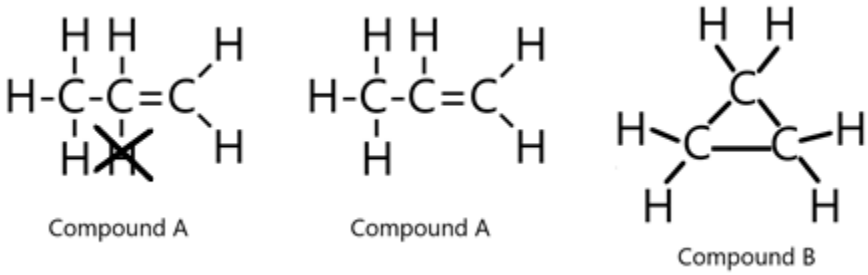
5. (**3 marks**) As far as thermodynamics is concerned define a closed system. Write a sentence! Next give an example of a closed system.

A closed system is any system in which energy can be transferred between the system and its surroundings but matter cannot.

Example: a can of beer, the cylinder of compressed air in the lab (when not in use, valve closed)

6.(**6 marks**) Calculate the enthalpy of combustion of propene compound A and cyclopropane compound B using the thermodynamic data given in the formulas and constants section. The final result should have the unit kJ/mol and should be based on the heat produced per mole of compound combusted. The two enthalpies for the isomers are different. Which compound is more stable and how can you tell?



I am sorry about my drawing error. Thanks for pointing it out during the exam Trevor!

I corrected this on the board but this information would not have reached students who were writing with CAL. See me if you feel that you lost marks because of this.

C_3H_6 (g) + 4.5O_2(g) → 3CO_2(g) + 3H_2O(l) Equation for both A and B (they are isomers)

$\Delta H° = \Sigma v_p \Delta H_f°$(products) – $\Sigma v_r \Delta H_f°$(reactants)

Compound A: $\Delta H° = \{(3\text{ mol})(-393.5\frac{kJ}{mol}) + (3\text{ mol})(-285.8\frac{kJ}{mol})\} - \{(1\text{ mol})(+20.42\frac{kJ}{mol}) + (4.5\text{ mol})(0\frac{kJ}{mol})\}$

$\Delta H° = -2058.32$ kJ for the reaction as written (combustion of one mol propene) so $-2058.32\frac{kJ}{mol}$

Compound B: $\Delta H° = \{(3\text{ mol})(-393.5\frac{kJ}{mol}) + (3\text{ mol})(-285.8\frac{kJ}{mol})\} - \{(1\text{ mol})(+53.3\frac{kJ}{mol}) + (4.5\text{ mol})(0\frac{kJ}{mol})\}$

$\Delta H° = -2091.2$ kJ for the reaction as written (raction of one mol cyclopropane) so $-2091.2\frac{kJ}{mol}$

Propene is more stable as it releases less energy when combusting (it stores less energy as a fuel). We can also see this from the enthalpy of formation which is less positive for propane (e.g. it takes less energy to make propane from the elements compared to cyclopropane which stores a greater amount of energy and is therefore less stable).

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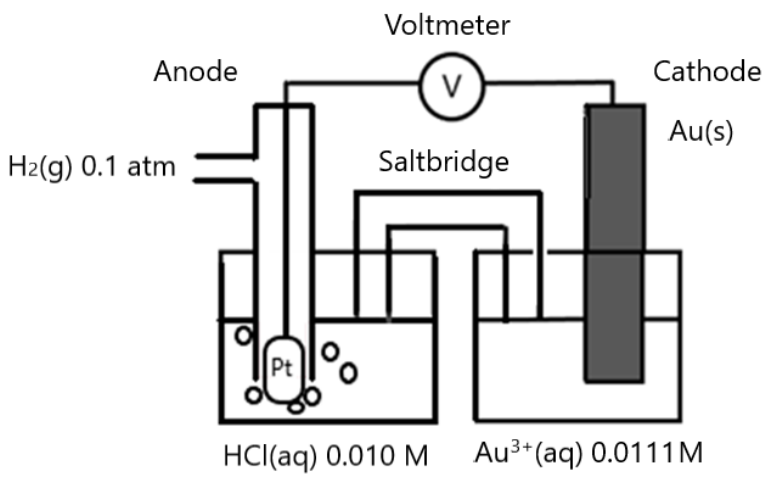
7. (**10 marks**) Draw a hydrogen electrode properly connected to a gold half cell. Name all vital parts of this Voltaic cell in your drawing. The hydrogen ion concentration in the hydrogen electrode is 0.010 M and the $H_2(g)$ pressure is 0.1 atm. The gold electrode is dipping into a solution with a gold(III) ion concentration of 0.0111 M.



Which electrode is the Anode and which electrode is the Cathode in your drawing?

The hydrogen electrode is anode (Pt) and the gold electrode is cathode

Give the half reactions that occur on both electrodes and give the overall cell reaction.

Cathode half reaction	**[$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$]x2**	**+ 1.52 V**
Anode half reaction	**[$H_2(g) \rightarrow 2H^+(aq) + 2e^-$]x3**	**0 V**
Overall cell reaction	**$2Au^{3+}(aq) + 3H_2(g) \rightarrow 2Au(s) + 6H^+(aq)$**	**$E_{cell}^o = + 1.52$ V**
	$2Au^{3+}(0.0111) + 3H_2(0.1) \rightarrow 2Au(s) + 6H^+(0.01)$	

Calculate the cell potential E_{cell}

$$E_{cell} = E_{cell}^\circ - \frac{0.0592\,V}{n}\log Q = +1.52\text{ V} - \frac{0.0592\,V}{6}\log\frac{[1]^2[10^{-2}]^6}{[0.0111]^2[0.1]^3} \qquad (Q = 8.116\times10^{-6})$$

E_{cell} = + 1.52 V – (-0.050) V

E_{cell} = + 1.57 V

Draw an arrow in your cell diagram indicating the direction flow of electrons: **Anode to Cathode**

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8. (**8 marks**) Calculate the pH of 0.111 M aqueous sodium nitrite: $NaNO_2$(aq). The acid ionization constant for nitrous acid (HNO_2) is **K_a= 7.2 x 10^{-4}**

1) Dissociate the ionic compound in water

   $NaNO_2$(s) ↔ Na^+(aq) + NO_2^-(aq)

   after complete dissociation the concentration of sodium and nitrite ions will be 0.111 M

2) Hydrolyze (react with water) the ion of the weak acid

	NO_2^-(aq) + H_2O(aq) →	**HNO_2(aq) +**	**OH^-(aq)** now set up ICE table
Initial	0.111 M	0 M	0 M
Change	-x M	+x M	+ x M
Equilibrium	(0.111 – x)M	x M	x M

We are given k_a for HNO_2 but its on the wrong side of the equation, we need k_b for the corresponding base **NO_2^-** (aq). Use **K_a x K_b = 10^{-14}** which yields k_b(**NO_2^-**) = $\frac{1x10^{-14}}{7.2\ x\ 10^{-4}}$ = 1.388 x 10^{-11}

1.388 x 10^{-11} = $\frac{[HNO_2][OH^-]}{[NO_2^-]} = \frac{x^2}{(0.111-x)}$

We can solve the quadratic equation or try the approximation if x << 0.111 then $\frac{x^2}{(0.111-x)} \sim \frac{x^2}{0.111}$ and

1.388 x $10^{-11} \sim \frac{x^2}{0.111}$ Which gives x = 1.242 x 10^{-6} = [OH^-] and pOH = 5.91

pH = 14- 5.91 1.242 x 10^{-6} << 0.111, our simplifying assumption is valid.

pH = 8.09 This is mildly basic which makes sense for the salt of a strong base with a weak acid.

Some of you combined step 1 and 2 in one which can be done. The equation would be:

	$NaNO_2$(s) + H_2O(l) →	**HNO_2(aq) +**	**NaOH(aq)**
Initial	0.111 M	0 M	0 M
Change	-x M	+x M	+ x M
Equilibrium	(0.111 – x)M	x M	x M

The ICE table and the follow up would be the same!

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Formulas and constants:

Selected standard electrode half potentials:

$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$	**+1.52 V**
$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$	**1.229 V**
$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	**+0.771 V**
$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$	**+ 0.340 V**
$2H^+(aq) + 2e^- \rightarrow H_2(g)$	**0 V (as defined)**
$Mo^{3+}(aq) + 3e^- \rightarrow Mo(s)$	**- 0.20 V**
$Cr^{3+}(aq) + e^- \rightarrow Cr^{2+}(aq)$	**- 0.424 V**
$Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$	**- 0.440 V**
$Mn^{2+}(aq) + 2e^- \rightarrow Mn(s)$	**- 1.18 V**
$Mg^{2+}(aq) + 2e^- \rightarrow Mg(s)$	**- 2.356 V**

Some useful equations: **$S = k\ln W$, $k = 1.38064852 \times 10^{-23}\ JK^{-1}$**

$\Delta H° = \Sigma v_p \Delta H_f°(products) - \Sigma v_r \Delta H_f°(reactants)$ **$\Delta G° = \Delta H° - T\Delta S°$**

Selected standard enthalpy of formations:

CH_3NO_2 (l)	**– 112.6 kJ/mol**
Cyclopropane C_3H_6 (g)	**+ 53.3 kJ/mol**
propene C_3H_6 (g)	**+ 20.42 kJ/mol**
CO_2(g)	**- 393.5 kJ/mol**
H_2O(l)	**-285.8 kJ/mol**

$E°_{cell} = E°_{reduction\ half\ cell} - E°_{oxidation\ half\ cell}$ (where E° refers to the reduction half potentials)

$$E_{cell} = E_{cell}° - \frac{0.0592\ V}{n} \log Q$$

$pH = -\log[H_3O^+] = -\log[H^+]$ **$pOH = -\log[OH^-]$** **$pH + pOH = 14 = pK_w$**

For an acid HA that ionizes in water to give H_3O^+ and A^- the acid constant K_a is defined as:

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

For an acid and its conjugated base (or a base and its conjugated acid) in water: **$K_a \times K_b = K_w = 10^{-14}$**

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